3 Bipolar Junction Transistors

BJT (BIPOLAR JUNCTION TRANSISTORS)

INTRODUCTION

- After the diodes, the most popular device in the semiconductor devices are the transistors.
- BJT (Bipolar Junction Transistor) is the first transistor invented in December 1947 by John Bardeen and Walter Brattain. The team was headed by William Shockley.
- It replaced the bulky vacuum tube and has evolved since then.
- It is used both in analogue as well as digital electronics.
- It is used in the amplification of weak signals and in switching operations also.

Grey Matter Alert!

The team who invented BJT also received a Nobel prize in the year 1956 for their work in introducing the first transistor in the electronics industry.

CONSTRUCTION AND WORKING

- Bipolar Junction Transistor is a three-layer semiconductor device consisting of either two n-type and one p-type layer of material or one n-type and two p-type layers of the material.
- Hence, there are two junctions, three layers, and three terminals in BJT.
- The BJT can be of two types:

1) NPN-BJT:

- In this transistor, the emitter and collector are n-doped, and the base is p-doped.
- The current is mainly due to electrons.

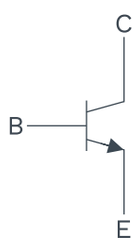


Fig. 3.1 Symbol of NPN-BJT

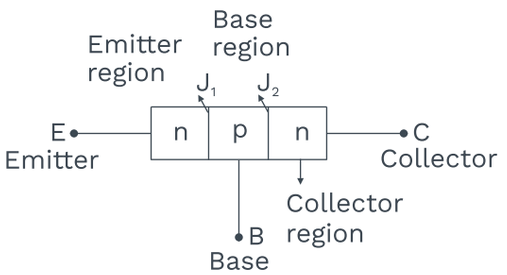


Fig. 3.2 a)

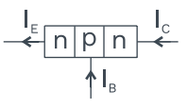


Fig. 3.2 b)

Fig. 3.2 Construction of NPN-BJT

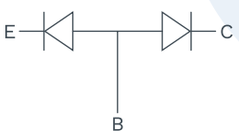


Fig. 3.3 NPN-BJT Using Diodes

2) PNP-BJT:

- In this transistor, the emitter and collector are p-doped and the base is n-doped. The current is mainly due to holes.

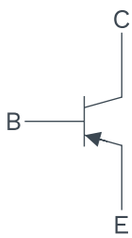


Fig. 3.4 Symbol of PNP-BJT

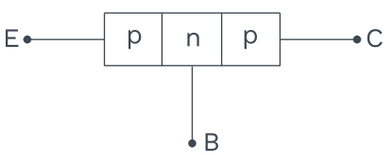


Fig. 3.5 a)

Grey Matter Alert!

John Bardeen is the only physicist who was awarded Nobel Prize in 2 years [1956 and 1972].

Grey Matter Alert!

The bifurcation of BJT:

- "Bipolar" because the conduction is due to both electrons and holes.
- "Junction" as there are 2 p-n junctions in the device.
- "Transistor" = TRANSFER + RESISTOR because it transfers the input signal current from low resistance to high resistance circuit.

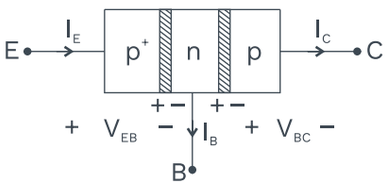


Fig. 3.5 b)

Fig. 3.5 Construction of PNP-BJT

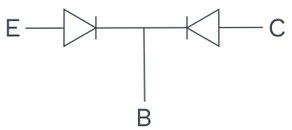


Fig. 3.6 PNP-BJT Using Diodes

- There are a few specific features that differentiate one region from other regions.

Emitter region:

- Emitter region emits or supplies the majority of carriers. In NPN transistor and PNP transistor, the emitter will supply electrons and holes, respectively.
- Emitter region is heavily doped. So, thus emitter can supply a large number of carriers.
- Impurities are added in the ratio of 1:10^3.

Collector region:

- Collector region is largest in size and moderately doped.
- Collector region receives or collects majority carriers coming from the emitter.
- In a BJT the collector junction develops a large amount of heat because it operates at higher voltage and higher current.
- If the collector region is lightly doped, then its conductivity will reduce, which is undesired for the transistor.
- If the collector region is heavily doped, then the breakdown voltage of the collector junction will decrease, which is also undesired. So, the collector is moderately doped so that it has better conductivity and the collector junction has a higher breakdown voltage.

Rack Your Brain

Can two practical diodes be connected back to back to work as a transistor?

Grey Matter Alert!

NPN transistor is superior to PNP transistor because the mobility of electrons is greater than the mobility of holes.

Base region:

- Region through majority of carriers travel from emitter to collector.
- Area of the base region is small and it is lightly doped.
- The recombination of electron, and hole inside the base will be reduced by keeping it narrow in width and by doping it lightly.
- Impurities added in the ratio $1:10^8$.

Normal mode-transistor action:

For a p-n-p transistor:

- The p+ or p region(heavily doped) is called the emitter; p region is called the collector and the centre n region is called the base.
- For a good p-n-p transistor, all the holes [majority carriers] injected by the emitter into the base should get collected.
- Hence, the n-type base is made very thin to reduce the recombination of electrons and holes.
- Emitter-Base Junction is forward biased.
- Total current is the sum of majority and minority carriers generated, but we want it to be only due to holes. Hence this is heavily doped. (by injection of majority carriers)
- n-type base is made very thin as possible so that we do not waste any injected carriers. (Here, the majority of carriers are electrons.)
- Both the depletion layers will penetrate more into the base region. This reduces the base width of the transistor.
- Recombination factor = 1.
- The p-type base-collector junction is reverse biased so that holes cannot go back to the base, and minority carriers from the base can go to the collector easily.

Grey Matter Alert!

Emitter junction- is named as “emitter” as it emits a large number of majority carriers to the collector.

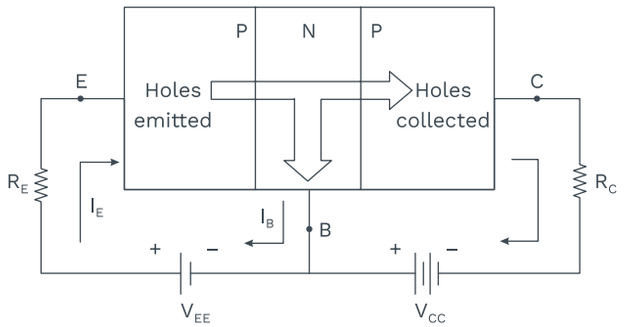


Fig. 3.7 Operation of PNP

For an n-p-n transistor:

- Holes are injected from base to emitter.
- Electrons are injected from emitter to base.
- Now, the emitter is heavily doped, so more electrons are injected.
- The base is lightly doped, so recombination in the base region is ideally negligible.

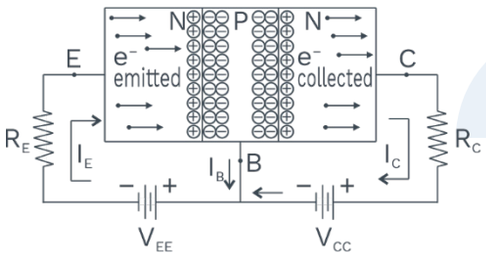


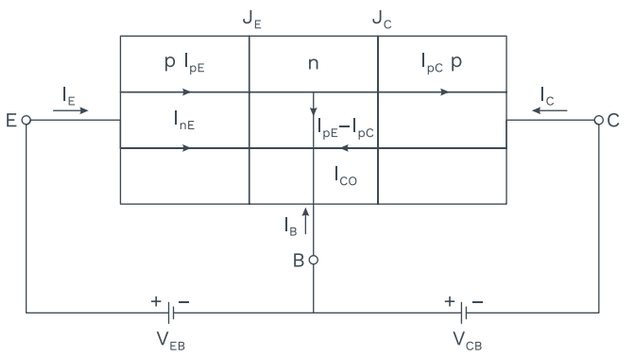
Fig. 3.8 Operation of NPN

TRANSISTOR ACTION

- The transistor action is defined by the amplification action, which takes place when the emitter-base junction is forward biased and collector-base junction is reverse biased.
- In the operation of the transistor, there are various current components involved.

Definitions

- α is called the current transfer ratio.
- β is called the base to collector current amplification factor or the base transport factor. It is the fraction of injected holes made across the base to the collector region successfully.



- The emitter current I_E consists of hole current I_{pE} due to holes crossing from emitter to base and electron current i_{nE} due to electron crossing from the base into the emitter.
- The ratio of the hole to electrons current, $\frac{I_{pE}}{I_{nE}}$ crossing the emitter junction is proportional to the ratio of the conductivity of p-material to that of n-material.
- Not all the holes crossing emitter junction I_E reaches collector junction J_C because some of them combine with electrons in n-type base.
- If I_{pC} is the hole current at J_C, there must be bulk recombination current $I_{pE} - I_{pc}$ leaving base, as shown above in figure.
- If $I_E = 0$ (emitter open-circuited), then I_{pC} would be zero. Under these conditions, base and collector would act as a reverse biased diode and collector current I_C would be equal to the reverse saturation current I_{co}.
- If $I_E \neq 0$, then from the figure.
  $I_C = I_{co} - I_{pc}$
- For p-n-p transistor, I_{co} consists of holes moving across J_C from left to right (base to collector) and electron crossing J_C in the opposite direction. Since the assumed reference direction for I_{co} in the above figure is from right to left. So, for the pnp transistor, I_{co} is negative, and for the npn transistor, I_{co} is positive.

Previous Years' Question

The DC current gain (β) of a BJT is 50. Assuming that the emitter injection efficiency is 0.995, the base transport factor is:

a) 0.980 **b)** 0.985
c) 0.990 **d)** 0.995

Emitter or injection efficiency (γ):
It is the ratio of injected carrier current at the emitter junction (j_E) to the total current.

$$\gamma = \frac{\text{Current of carriers at } J_E \text{ due to carriers of the emitter}}{\text{Total emitter current}}$$

In the case of the p-n-p transistor,

$$\gamma = \frac{I_{pE}}{I_E} = \frac{I_{pE}}{I_{pE} + I_{nE}}$$

Where I_{pE} = Injected hole diffusion current at emitter junction (j_E)
I_{nE} = Injected electron diffusion current at emitter junction (J_E)

Transport factor (β^*):
It is defined as the ratio of injected carrier current reacting J_C to the injected carrier current at J_E.

$$\beta^* = \frac{\text{Injected carrier current reaching } J_C}{\text{Injected carrier current at } J_E}$$

In the case of the p-n-p transistor, we have

$$\beta^* = \frac{I_{pC}}{I_{pE}}$$

Large signal current gain (α):
The Large signal current gain of a common base transistor is defined as the ratio of the negative of the collector-current increment to the emitter-current change from zero (cut-off) to I_E.

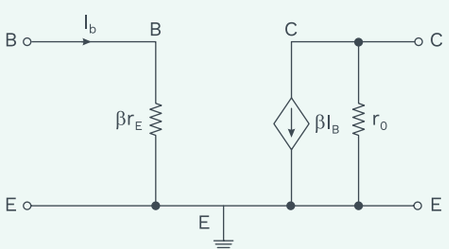
$$\alpha = \frac{-(I_C - I_{CO})}{I_E}$$

Since I_C and I_E have opposite signs, α is defined as always positive.

- For the typical numerical value of α lies in the range of 0.90 to 0.995.

Previous Years' Question

For a BJT, the common base current gain α = 0.98 and the collector-base junction reverse bias saturation current I_{CO} = 0.6 μA. This BJT is connected in the common emitter mode and operated in the active region with a base drives current I_B = 20 μA. The collector's current I_C for this mode of operation is



a) 0.98 mA **b)** 0.99 mA
c) 1.0 mA **d)** 0.01 mA

Relationship between α, β^* and γ:

$$\alpha = \frac{-(I_C - I_{CO})}{I_E} = \frac{I_{pC}}{I_E}$$

$$\alpha = \frac{I_{pC}}{I_{pE}} \times \frac{I_{pE}}{I_E}$$

$$\alpha = \beta^* \gamma$$

COLLECTOR CURRENT (IN ACTIVE REGION)

If the transistor is in the active region, then the collector is given by

$$I_C = -\alpha I_E + I_{CO} \quad \ldots(i)$$

If currents I_E, I_B and I_C are assumed to be following into the transistor then:

$$I_E + I_B + I_C = 0$$

$$I_E = -(I_B + I_C) \quad \ldots(ii)$$

From equations i) and ii), we get:

$$I_C = \alpha(I_B + I_C) + I_{CO}$$

$$I_C - \alpha I_C = \alpha I_B + I_{CO}$$

$$I_C(1 - \alpha) = \alpha I_B + I_{CO}$$

$$I_C = \frac{\alpha}{1-\alpha} I_B + \frac{I_{CO}}{1-\alpha}$$

Let, $\frac{\alpha}{1-\infty} = \beta, 1 + \beta = 1 + \frac{\alpha}{1-\alpha} = \frac{1}{1-\alpha}$

$$I_C = \beta I_B + (1+\beta) I_{CO}$$

Where β = common emitter large signal current gain

α = common base large signal current gain

$$\beta = \frac{\alpha}{1-\alpha}$$

Previous Years' Question

In a uniformly doped BJT, assume that N_E, N_B and N_C are the emitters, base and collector doping in atoms/cm^3, respectively. If the emitter injection efficiency of the BJT is close to unity, which one of the following conditions are TRUE?

a) $N_E = N_B = N_C$
b) $N_E >> N_B$ and $N_B > N_C$
c) $N_E = N_B$ and $N_B < N_C$
d) $N_E < N_B < N_C$

$$(1-\alpha)\beta = \alpha$$

$$\alpha = \frac{\beta}{1+\beta}$$

A transistor with large β will have an excellent amplifier.
If $\alpha \to 1$, then $\alpha \to \infty$ has to fabricate a transistor with large β, α should be close to 1 but $\alpha = \beta^{*}\gamma$.
Therefore to have a large β, the following conditions should be satisfied in a BJT.

1) $N_E >> N_B$

Where N_E = Doping concentration in the emitter

N_B = Doping concentration in base
If $N_E >> N_B$, the γ becomes very close to 1.

2) $W_B < L_B$
Where W_B = base width
L_B = diffusion length
If $W_B << L_B$ then β^{*} becomes very close to 1.

I_{CBO}: It is the current flowing from collector to base when the emitter is open-circuited.

$I_{CBO} = I_{CO}$

I_{CEO}: It is the current flowing from collector to emitter when the base is open circuited.

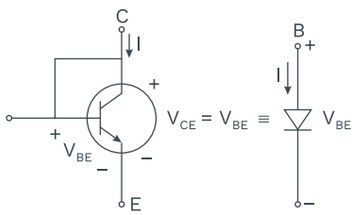
$$I_{CEO} = (1+\beta)I_{CO} = \frac{I_{CO}}{1-\alpha}$$

$$I_{CEO} = (1+\beta)I_{CBO} = \frac{I_{CBO}}{1-\alpha}$$

- When the collector is shorted to be base (i.e. $V_{CB} = 0$) then BJT will behave like a diode. (Single junction device) because current passing through BJT will be decided by the voltage V_{BE} across J_E. In this case, current through diode connected transistor will be equal to the current passing through J_E.

$$I = I_{EO}e^{\frac{V_{BE}}{\eta V_T}}$$

where I_{EO} = reverse saturation current of J_E.



Example 1:
For a BJT collector current of 0.9mA, the base current is 20μA. Calculate α.

Sol:

$I_c = 0.9mA$

$I_B = 20\mu A$

$\therefore \quad \beta = \frac{\alpha}{1-\alpha}$

$\beta(1 - \alpha) = \alpha$

$\beta = (1 + \beta)\ \alpha$

$\alpha = \frac{\beta}{1+\beta}$

$\alpha = \frac{I_c}{I_E}$

$\alpha = \frac{0.9m}{0.9m + 20\mu}$

$\therefore \quad \alpha = 0.978$

Example 2:
Given the common base current gain of 0.98, Calculate the common emitter current gain.

Sol:
Given, : $\alpha = 0.98$

By: $\beta = \frac{\alpha}{1-\alpha}$

$\therefore \quad \beta = \frac{0.98}{0.02}$

$\beta = 49$

Example 3:
A transistor has $\beta = 89$, and emitter injection efficiency is 0.998. Find the base transport factor.

Previous Years' Question

Choose the correct match for the input resistance of various amplifier configurations shown below.

Configuration	Input resistance
CB : Common Base	LO : Low
CC : Common Collector	MO :Moderate
CE : Common Emitter	HI : High

a) CB-LO, CC-MO, CE-HI
b) CB-LO, CC-HI, CE-MO
c) CB-MO, CC-HI, CE-LO
d) CB-HI, CC-LO, CE-MO

Sol:

Given,

$\beta = 89$

$\gamma = 0.998$

We know that,

$\alpha = \beta^* \gamma$

$\beta^* = \frac{\alpha}{\gamma}$

$\alpha = \frac{\beta}{1+\beta} = \frac{89}{1+89} = 0.988$

$\beta^* = \frac{0.988}{0.998} = 0.989$

Example 4:

A transistor has $I_B = 20\mu A$, $I_{CBO} = 200nA$ and $\beta=100$. Calculate common emitter current gain, emitter current and I_{CEO}.

Sol:

Given, $I_B = 20\,\mu A, I_{CBO} = 200\,nA$

$\beta = 100$

Common emitter current gain:

$\alpha = \frac{\beta}{1+\beta}$

$\alpha = \frac{100}{1+100}$

$= \frac{100}{101}$

$= 0.99$

Collector current:

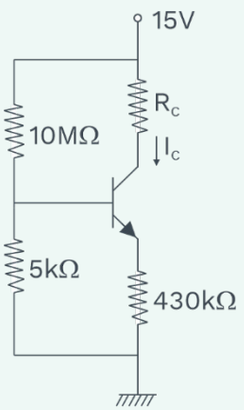
$I_C = \beta I_B + (1+\beta) I_{CBO}$

$I_C = 100 \times 20 \times 10^{-6} + 101 \times 200 \times 10^{-9}$

$I_C = 2000 \times 10^{-6} + 20.2 \times 10^{-6}$

Previous Years' Question

In the circuit of the figure, assume that the transistor is in active region. It has a large β and its base-emitter voltage is 0.7 V. The value of I_C is:



a) indeterminate since R_c is not given.
b) 1 mA
c) 5 mA
d) 10 mA

$I_C = 2020.2 \times 10^{-6} = 2020.2\,\mu A$

$I_C = 2.02 mA$

$I_E = -I_B - I_C$

$I_E = -0.02 - 2.02$

$I_E = -2.04\,mA$

$I_{CEO} = (1+\beta) I_{CBO}$

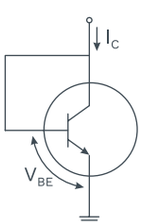
$= 101 \times 200 \times 10^{-9}$

$= 20.2\mu A$

Example 5:

For the given transistor shown below, the reverse saturation current of the junction in the room temperature (300K) is 10^{-16} A, the emitter current is-

Given,: $V_{BE} = 0.7\,V$



Sol:

Given,

$I_0 = 10^{-10} A\ ;\ V_{BE} = 0.7\,V$

When two terminals of a transistor are shorted, it acts as a diode.

$$I = I_0 \left(e^{\frac{V_{BE}}{V_T}} - 1 \right)$$

$$I = 10^{-16} \left(e^{\frac{0.7}{26 \times 10^{-3}}} - 1 \right)$$

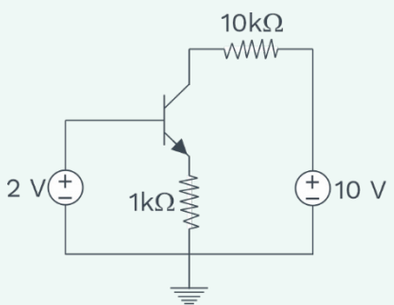
$= 48.26\,\mu A$

Grey Matter Alert!

V_{CE} (sat) for different transistors-
NPN Ge-based=0.1V
NPN Si-based=0.2V
PNP Ge-based=-0.1V
PNP Si-based=-0.2V

Previous Years' Question

For the BJT circuit sown, assume that the β of the transistor is very large and V_{BE}=0.7V. The mode of operation of the BJT is:



a) cut-off
b) saturation
c) normal active
d) reverse active

MODES OF OPERATION

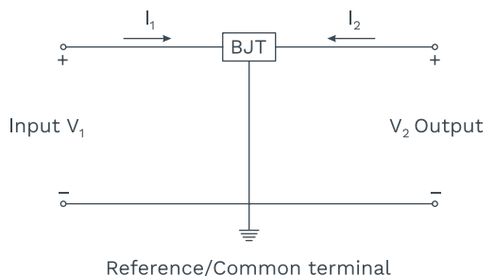
Depending upon the polarity of the biasing voltage, we have different modes of operation of transistors, and each mode has a different application. Different modes of operation of BJT shown in the table below.

Mode	Emitter-Base Junction	Collector-Base Junction	Properties	Application
Cut-off	Reverse bias	Reverse bias	Very high internal resistor	OFF switch
Active	Forward bias	Reverse bias	Excellent transistor action	Amplifier
Saturation	Forward bias	Forward bias	Very low internal resistance	ON switch
Reverse active	Reverse bias	Forward bias	Very poor transistor	Attenuator (Practically not used)

Table 3.1 Modes of Operation

CONFIGURATIONS OF BJT

BJT is three terminal device through which the input is applied, and output is taken across the other two terminals as shown below.



- There are three possible configurations of BJT. In every configuration, one of the terminals out of the three terminals (base, emitter, collector) acts as a reference terminal, one as input and the other as output.
- Common emitter configuration

Grey Matter Alert!

After the early effect

i) Collector current, α and β of the transistor increases.

ii) Recombination current & base current reduces. ($\therefore I_C = \alpha I_E$)

- Common base configuration
- Common collector configuration
- The internal mechanism of current flow remains the same in all the three configurations.
- We keep an assumption while plotting BJT characteristics, input current I_1 and output voltage V_2 as to be independent parameters, whereas input voltage V_1 and output current I_2 are considered as dependent parameters.

Thus, $V_1 = f_1(I_1, V_2)$

$V_2 = f_2(I_1, V_2)$

If V_2 = constant, then

$V_1 = f_1(I_1)$

Now, if a graph is plotted between the input current and input voltage constant, then it is known as input characteristics.

If I_1 = constant, then

$I_2 = f(V_2)$

Now, if a graph is plotted between the output current and output voltage keeping input current constant, then it is known as output characteristics.

1) Common emitter amplifier:

- For common emitter configuration: I_c is the output current, I_B is the input current and V_{CE} is the output voltage.

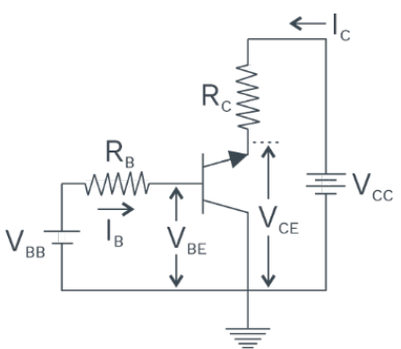


Fig. 3.9 a) Common Emitter Amplifier (For p-n-p BJT)

Previous Years' Question

The early effect of a bipolar junction transistor is caused by:

a) fast turn-on.
b) fast turn-off.
c) large collector-base reverse bias.
d) large emitter-base forward bias.

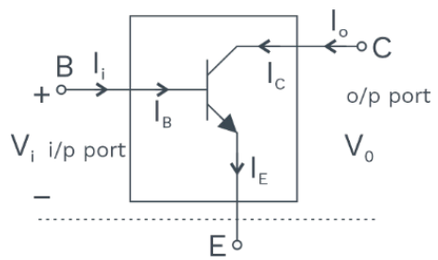


Fig. 3.9 b) Common Emitter Amplifier (For n-p-n BJT)

- So here, if we analyse a B.J.T. as a connection of 2 diodes connected back to back, we will say that one of the diodes will be the input acting diode with a junction formed between the emitter and base junction.

Current and voltage equations:

- $V_i = V_{BE}$

($V_i = 0$ in ideal case which is equal to 0.5 or 0.7 for Silicon practically)

- $I_i = I_B$

$V_0 = V_{CE}$(V_0 = any positive voltage)

$I_O = I_C$

- For Ideal transistor: $\alpha = 1$

$I_C + I_B = I_E$

then, $I_i = I_B = 0$

$I_C = I_o = I_E$

Input characteristics:

- For B.J.T to operate in active mode, we need to keep the emitter and base junction forward biased. So, for a particular value of output voltage (V_{CE}), the input characteristics will be the same as for a p-n junction diode under forward biased conditions.
- But in input characteristics, we have to look at the effect of V_{CE} on the input current I_B also, and here early effect comes into the picture.

Previous Years' Question

The phenomenon's known as "Early Effect" in a bipolar transistor refers to a reduction of the effective base-width caused by:

a) electron-hole recombination at the base.

b) reverse biasing of the base-collector junction.

c) forward biasing of emitter-base junction.

d) early removal of stored base charge during saturation-to-cutoff switching.

- We know that base is lightly doped due to which when the collector-base junction gets reverse biased for B.J.T. to work in active mode, the width of depletion layer associated with collector-base junction gets increased due to reverse biasing, which decreases the width of effective region of the base where the electrons coming from the emitter region tends to combine with the holes present in p-type of the base, due to which I_B decreases.

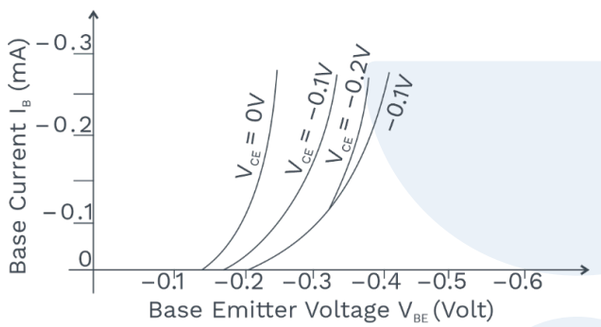


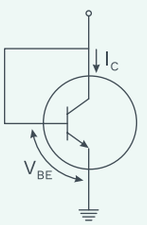
Fig. 3.10 Input Characteristics of Common Emitter Amplifier

Output characteristics:

- It is the curve between the output current and output voltage for a given configuration for different values of input current for that configuration.
- So, for common emitter configuration, output characteristics are the curve between I_c(y-axis) and V_{CE}(x-axis) for different values of I_B.
- To draw output characteristics for a particular value of I_B, we can consider B.J.T. as a connection of 2 diodes connected back to back and then we can consider the second diode, which is having collector base junction.
- Now, since this junction is reverse biased for a B.J.T. to work in an active mode, the output characteristics under the common emitter configuration will be the same as for a p-n junction diode under reverse biased condition.

Previous Years' Question

For an n-p-n transistor connected as shown in the figure, V_{BE} = 0.7 volts. Given that reverse saturation current of the junction at room temperature, 300°K is 10^{-13} A, the emitter current is:



a) 30 mA **b)** 39 mA
c) 49 mA **d)** 20 mA

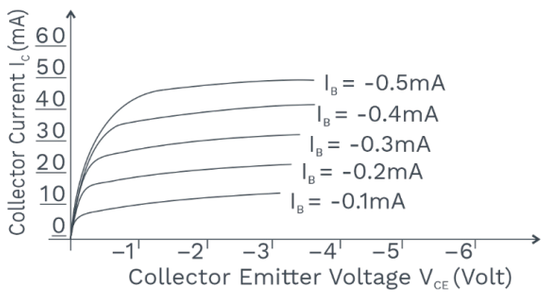


Fig. 3.11 Output Characteristics of Common Emitter Ampli ier

Active region:

In this region, the collector junction is biased in reverse direction and the emitter junction is in forward biased direction.

- In the active region, the transistor output current is more sensitive to an input signal.
- BJT is used as an amplifier in an active region. This region is most preferable for the operation of transistor.

Saturation region:

In this region, collector junction as well as emitter junction is forward biased.

- In this region, bottoming occurs, $|V_{CE}|$ drops to few tenths of a volt, and the collector current is approximately independent of the base current, for given values of V_{CC} and R_L.
- BJT is used as ON Switch in this region.

Cut-off region:

In this region, both collector junction and emitter junction are reverse biased.

The region below I_B = 0μA is Cut-off region, however we now find that appreciable collector current may exist under these conditions.

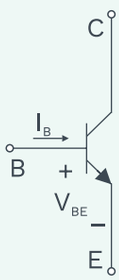
If $I_B = 0$, then $I_E = -I_C$, hence we have-

$$I_C = -I_E = \frac{I_{CO}}{1-\alpha} = I_{CEO}$$

The actual collector current with the collector junction reverse biased and base open-circuited is denoted by the symbol I_{CEO}.

Previous Years' Question

If transistor in the figure is in saturation, then:



β_{dc} denotes the dc current gain

a) I_C is always equal to $\beta_{dc} I_B$

b) I_C is always equal to $-\beta_{dc} I_B$

c) I_C is greater than or equal to $\beta_{dc} I_B$

d) I_C is less than or equal to $\beta_{dc} I_B$

- Since, even in the neighbourhood of Cut-off, α may be as large as 0.9. For germanium then $I_C \simeq 10I_{CO}$ at $I_B = 0$.
- Accordingly, in order to cut off the transistor, it is not enough to reduce I_B to zero, instead it is necessary to reverse bias the emitter junction slightly.
- BJT is widely used as an off switch in this region.

Note: There are two methods of determining of saturation region of transistor.

- If I_C and I_B can be determined independently from the circuit under consideration, the transistor is in saturation.

If $|I_B| \geq \frac{I_C}{h_{FE} \text{or} |\beta|}$

If V_{CB} is determined from the circuit configuration and if this quantity is positive for a p-n-p transistor (or negative for n-p-n transistor), transistor is in saturation, of course; the emitter junction must be simultaneously forward biased, but then we should not be testing for saturation if this condition was not satisfied.

Properties:

- The input resistance is moderate (around 1 kΩ).
- The output resistance is moderate (50 kΩ to 500 kΩ).
- Current gain is moderate (Typical value 49).
- Voltage gain is moderate.
- The power gain is highest (Typical value 4226).
- The input voltage and output voltage are out of phase, i.e., Phase shift = 180°.

Application:

- It is the most common and frequently used amplifier circuit.

Current gain (β):

The current gain is defined as the ratio of the output current to the input current. It is denoted by β.

In a common emitter, the output current is I_C, and the input current is I_B.

$$\text{Current gain} = \beta = \frac{\text{output current}}{\text{Input current}} = \frac{I_C}{I_B}$$

For AC situation, an AC beta is defined as-

$$\beta_{AC} = \left.\frac{\Delta I_C}{\Delta I_B}\right|_{VCE=\text{constant}}$$

- The range of β is 20 to 500.
- β is called the current gain of transistor in common emitter mode.

- β is also denoted by h_{FE}.
- $\beta_{DC} > \beta_{AC}$ or $h_{FE} > h_{fe}$.

Effect of temperature on β-

We know that $\beta = \frac{I_C}{I_B}$

- The average time a carrier takes to travel from emitter junction through the base region to collector junction is called transit time 'τ_t'.

i.e. $\tau_t = \frac{W_B^2}{2D_B}$ ...(i)

where τ_t = Transit time
W_B = Base width
D_B = Diffusion constant.

If, in average time, one hole moves from the emitter junction to collector junction in τ_t seconds.

$\therefore$ In one second, from emitter to collector junction holes moves are $\frac{1(\tau_t)}{\tau_t}$ holes.

Holes that reach the collector junction will form 'I_C'.

$\therefore$ $I_C \propto \frac{1}{\tau_t}$ ...(ii)

The average time, a hole or electron will exist before recombination is called lifetime.

τ_p = Life times of holes
τ_h = Life times of electrons

In average time, one hole-electron recombination occurs in τ_p seconds.

$\therefore$ In one second $\frac{1}{\tau_p}$ recombination can happen.

Hence, the base should be supplied with $\frac{1}{\tau_p}$ electrons in one second to keep it neutral and there electrons will form base current, so:

$I_B \propto \frac{1}{\tau_p}$ ...(iii)

From equations ii) and iii), we get:

$\beta = \frac{I_C}{I_B} \cong \frac{\tau_p}{\tau_t}$

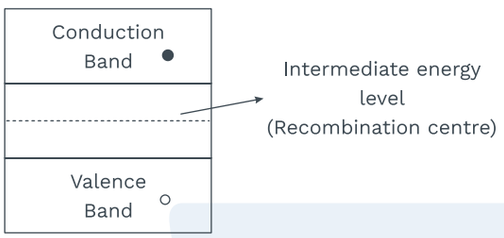
Grey Matter Alert!

Comparison of BJT and FET

BJT	FET
BJT is an asymmetrical device.	FET is a symmetrical device.
Emitter acts as a source in the BJT.	Source is analogous to emitter in BJT.
Base controls the current in BJT.	Gate controls the drain current in FET analogous to BJT.
Collector drains off the carriers in BJT.	Drain is analogous to collector in BJT.
Types of BJT are called: NPN Transistor PNP Transistor	Types of FET are called: N-channel FET P-Channel FET

- $W_B \downarrow \Rightarrow \tau_t \uparrow \Rightarrow \beta \uparrow$
- $T \uparrow \Rightarrow \tau_p$ or $\tau_n \uparrow \Rightarrow \beta \uparrow$

  When temperature increases, carrier lifetime increases, so β increases.
- In indirect semiconductor (Si or Ge), during recombination, free electrons moves into a hole or vacancy present in the valance band.



- When temperature increases, the electrons which are moved into the intermediate level will come back to the conduction band. This process is called thermal re-excitation.
- Thermal re-excitation will prevent recombination, and thereby lifetime increases, and as a result, β also increases.

2) Common base amplifier:

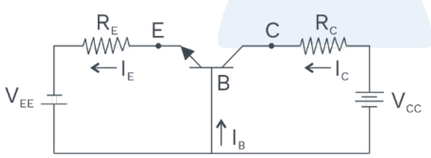


Fig. 3.12 Common Base Amplifier

- Common base terminology is derived from the fact that the:
  1) base is common to both input and output of the configuration.
  2) base is usually the terminal closest to or at ground potential.
- All current directions will refer to conventional (hole) flow, and the arrows in all electronic symbols have a direction defined by this convention.
- Note that the applied biasing (voltage sources) are such as to establish a current in the direction indicated for each branch.
- To describe the behaviour of common base amplifiers requires two sets of characteristics:
  1) Input or driving point characteristics.
  2) Output or collector characteristics

Input characteristics:

- We can consider the transistor as a connection of 2 diodes connected back to back. So to study input or driving point characteristics, if we take an n-p-n, active mode transistor; the diode connection will be of a forward biased n-p diode and a reverse biased p-n diode.
- To analyse input characteristics, we will infer to diode n-p diode which is clearly a forward biased diode, so the driving point characteristics correspond to characteristics of a forward bias diode.
- The base width modulation in Common base configuration plays a role in the input characteristics on increasing $V_{CB.}$

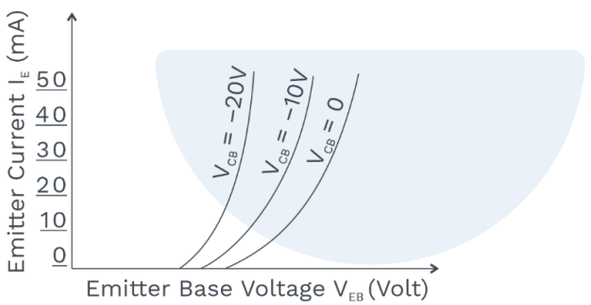


Fig. 3.13 Input Characteristics of Common base Amplifier

Output characteristics:

The output characteristics have three basic regions:

- Active region - defined by the biasing arrangements
- Cut off region - the region where the collector current is 0 Amperes.
- Saturation region- the region of the characteristics to the left of V_{CB} = 0 volts

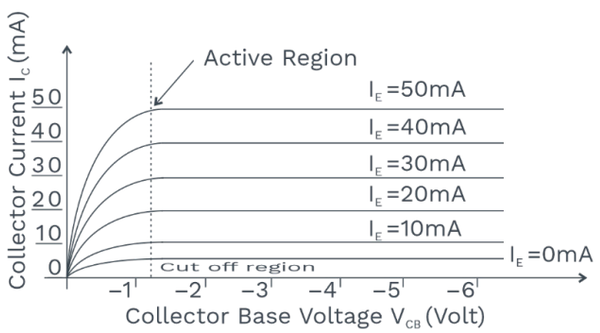


Fig. 3.14 Output Characteristics of Common base Amplifier

Active region: In this region collector junction is reverse biased and emitter junction is forward biased.

- If $I_E = 0$, then I_C is small and equals the reverse saturation current I_{CO} of collector junction is considered as a diode.
- Now, if forward emitter current caused to flow in the emitter circuit, then a fraction $-\alpha I_E$ of this current will reach the collector.

  $I_C = -\alpha I_E + I_{CO}$
- In the region, the collector current is independent of collector voltage and depends only upon emitter current. However, due to the early effect, small increases in the magnitude of collector current (I_C) with $|V_{CB}|$.

Saturation region:
In this region to the left of the ordinate $V_{CB} = 0$, and above the $I_E = 0$ characteristics in which both emitter and collector junctions are forward biased is called saturation region. Whenever the transistor enters into the saturation region, it can be used as an ON switch.

Cut-off region:
When both junction's collector and emitter are reverse biased is said to be in the cut-off region.
The region below and to the right of $I_E = 0$ characteristics, for which the emitter and collector junctions are both reverse biased.
In the cut-off region BJT used as an off switch.

Properties of CB configuration:
- It has very low input impedance ($R_i < 100\Omega$).
- It has high output impedance ($R_o > 1\ M\Omega$).
- It has the lowest current gain ($\alpha < 1$).
- It has the highest voltage gain.
- It has medium power gain (Typical value 68).
- The output and input voltages are in phase.
  i.e. Phase shift = 0^0
- In this configuration current gain is less. So, bandwidth is large hence C_B amplifier is widely used as a high-frequency amplifier.

Applications:
- Used as a high-frequency amplifier.
- Used as a non-inverting amplifier.
- Used as an impedance matching device between low resistance to high resistance.

Current gain (α):
It is defined as the ratio of the output current to the input current.
In common base, the output current is I_C and the input current is I_E.

$$\text{Current gain} = \frac{\text{Output Current}}{\text{Input Current}}$$

$$\alpha = \frac{I_C}{I_E}$$

The current gain of the common base amplifier is always less than unity.
i.e. $\alpha < 1$.

- For practical devices α typically extends from 0.90 to 0.998.
- For ac situations where the points of operation move on characteristics curve, an ac alpha is defined by:

$$\alpha_{AC} = \left.\frac{\Delta I_C}{\Delta I_E}\right|_{V_{CB}=\text{constant}}$$

- The α is called a common base amplification factor or current gain of common base transistor.

3) Common collector amplifier:

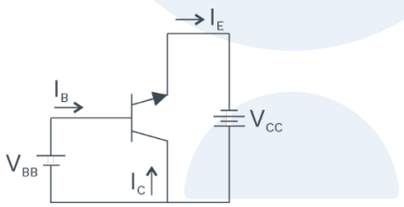


Fig. 3.15 Common-Collector Amplifier

- In a common collector configuration, the collector terminal is kept grounded, and it is the terminal which is common to both input and output circuit.
- Here, we generally talk about the output characteristics only rather than the input characteristics. It is a plot between output current I_E on the y-axis and output voltage V_{CE}, for different values of input current I_B.
- So, it's a curve between I_E and V_{CE} for various level of I_B.
- We know that $I_C = \alpha I_E$ and α approximates 1; hence, $I_c = I_E$
- So, for all practical reasons:

Output characteristics of common-collector configuration = output characteristics of common emitter configuration.

Properties of CC configuration:

- It has the highest input resistance (50 kΩ to 500 kΩ).
- It has the lowest output resistance (< 100 Ω).
- It has the highest current gain.

- The voltage gain is less than unity or close to unity.
- It has the lowest power gain (Typical value 48).
- Common collector configuration is also known as an emitter follower.
- Emitter follower is basically a current controlled voltage source (CCVS).
- The output voltage and input voltage are in phase. i.e. Phase Shift = 0°.

Applications:

- Used as a buffer (impedance matching device between high resistance to low resistance).
- Used as an audio frequency power amplifier.
- Used as a "boot step emitter follower".
- Used in designing voltage sweep circuits.

Current gain (γ):

It is defined as the ratio of output current to the input current.

The common collector, the output current is I_E, and the input current is I_B.

$$\text{Current gain } = \gamma = \left|\frac{I_E}{I_B}\right| = \left|\frac{I_C + I_B}{I_B}\right| = 1 + \beta$$

S. No.	Characteristics	CB configuration	CE configuration	CC configuration
1)	Input resistance	Very low (40Ω)	Low (50Ω)	Very high (750 kΩ)
2)	Output resistance	Very high (1 MΩ)	High (10kΩ)	Low (50Ω)
3)	Current gain	Less than unity	High (100)	High (100)
4)	Voltage gain	Small (150)	High (500)	Less than unity
5)	Leakage current	Very small	Very large	Very large
6)	Applications	For high frequency application	For audio frequency application	For impedance matching
7)	Phase shift between input and output	0°	180°	0°

Table 3.2

Relationship between α and β:

We know that $\beta = \frac{I_C}{I_B}$

$$\beta = \frac{I_C}{I_E - I_C}$$

$$\beta = \frac{1}{\frac{I_E}{I_C} - 1} = \frac{1}{\frac{1}{\alpha} - 1}$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\beta(1-\alpha) = \alpha$$

$$\alpha(1+\beta) = \beta$$

$$\alpha = \frac{\beta}{1+\beta}$$

Relationship between α, β and γ:

We know that,

$$\gamma = \frac{I_E}{I_B} = \frac{I_B + I_C}{I_B} = 1 + \frac{I_C}{I_B}$$

$$\gamma = 1 + \beta = 1 + \frac{\alpha}{1-\alpha}$$

$$\gamma = 1 + \beta = \frac{1-\alpha+\alpha}{1-\alpha}$$

$$\gamma = 1 + \beta = \frac{1}{1-\alpha}$$

Note: In a transistor, if various current gains are arranged in the ascending order, then the correct sequence is:

$$\alpha < \beta < \gamma$$

TEMPERATURE DEPENDENCE ON BJT PARAMETERS

i) I_{CO} V$_S$. Temperature

$$I_C = \beta I_B + (1+\beta) I_{CO}$$

$$I_C = I_O e^{\frac{V_{BE}}{V_T}}$$

$$I_C = \beta I_B$$

I_{CO} doubles for every 10°C rise in temperature.

$$T\uparrow \Rightarrow I_{CO} \uparrow \Rightarrow I_C \uparrow$$

ii) V_{BE} versus temperature

$$\frac{\partial V_{BE}}{\partial T} = -2.5\,mV\ /^{\circ}\ C$$

$$T\uparrow \Rightarrow V_{BE} \downarrow \Rightarrow I_C \downarrow$$

iii) β versus temperature

We know that,

$$\beta = \frac{I_C}{I_B}$$

$$T\uparrow \Rightarrow I_{CO} \uparrow \Rightarrow I_C \uparrow \Rightarrow \beta \uparrow$$

Its dependency is neglected because it is not a primary dependency on temperature.

Example 6:

In a transistor, when the emitter current is increased from 4 mA to 8 mA, the collector current increases from 3 mA to 6.8 mA. The value of β of transistor is

Sol:

Given,

$$\Delta I_E = 8mA - 4mA = 4mA$$

$$\Delta I_C = 6.8mA - 3mA = 3.8mA$$

We know that,

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

$$\Delta I_B = \Delta I_E + \Delta I_C$$

$$= 4mA - 3.8mA$$

$$= 0.2mA$$

$$\beta = \frac{3.8m}{0.2m} = 19$$

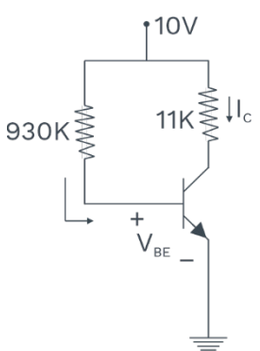
Example 7:
In the common-emitter configuration, the quiescent base current is determined by:
a) Base resistor
b) Load resistor
c) Both load resistor and base resistor
d) None

Sol:
In the common–emitter configuration the quiescent base current is determined by both load resistor and base resistor.
Hence, the correct option is c).

Example 8:
Which transistor configuration is most suitable for impedance matching?
a) CB Configuration
b) CE Configuration
c) CC Configuration
d) CE and CB Configuration

Sol:
The emitter follower configuration is mostly used as a voltage buffer these configurations are widely used in impedance matching applications because of their input impedance.
Hence, the correct option is c).

Example 9:
Consider the circuit in the figure below:
Assume: $\beta = 100$; Find I_C



Sol:

$$I_B = \frac{10 - 0.7}{930K} = 10\,\mu A$$

$I_C = 100 \times 10 \times 10^{-6} = 1mA$

Consider

$I_C R_C = 1 \times 10^{-3} \times 11 \times 10^3 = 11Volts$;

which is greater than V_{CC},

hence $I_C = 1mA = \beta I_B$ is not possible.

$I_C \neq \beta I_B$

And $I_C < 1\ mA < \beta I_B \rightarrow$ that is, the transistor will be in saturation.

EARLY EFFECT and PUNCH THROUGH

EARLY EFFECT

- In this phenomenon, the narrowing of the base of the transistor takes place.
- V_E is called the early voltage.
- V_{CB} increases, depletion width increases; hence the effective base width reduces.
- So, the early effect is the phenomenon when the magnitude of collector voltage increases and the effective base width of the transistor is reduced.
- As the base width is reduced, the base transit time is also decreased, and the BJT becomes faster in operation.
- More charge carriers reach the collector terminal as the recombination in the base decreases.

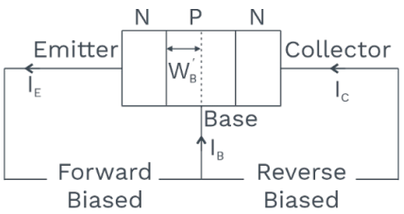


Fig. 3.19 Early Effect

- Where $\mathbf{W'_B}$ is the effective base width of the transistor

PUNCH THROUGH

- The process in which, by applying a larger collector voltage, the effective base width of the transistor reduces to zero is known as punch through or reach through.
- The collector becomes short circuit, and a very large current flows in the semiconductor destroying the transistor.

- **Breakdown voltage (V_{BR}):**

$$V_{BR} \propto \frac{1}{\text{Doping concentration}}$$

This collector breakdown voltage is always greater than the emitter junction breakdown voltage.

$BV_{C-B} > BV_{E-B}$

- So, when the collector junction voltage exceeds the collector breakdown voltage, the transistor is destroyed.

EBER MOLL'S MODEL

- In BJT, the Eber-Moll's model is quite complex, but it is valid in all the regions of BJT operation.

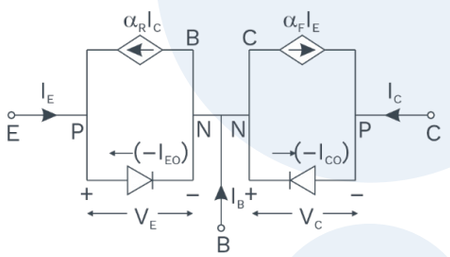


Fig. 3.20 Eber Moll's Model for p-n-p Transistor

- Two ideal diodes are connected back to back with reverse saturation currents. Two dependent current controlled current sources are shunting the ideal diodes.

In the above figure:

- α_R = common base current gain in inverse active mode
- α_F = common base current gain in forward active mode
- I_{EO} = reverse saturation current of the base-emitter junction
- I_{CO} = reverse saturation current of the base-collector junction

Current equations:

1) $$I_B = -\alpha_F I_E - I_{CO}\left(e^{\frac{V_E}{V_T}} - 1\right)$$

2) $$I_E = -\alpha_R I_C - I_{EO}\left(e^{\frac{V_E}{V_T}} - 1\right)$$

3) $I_E + I_C + I_B = 0$

$\Rightarrow I_B = -(I_E + I_C)$

From these equations, we can determine the mode of operation of the transistor:

- **Cut-off mode:**

  $\alpha_R I_C = \alpha_F I_C = 0$

So, $I_E \simeq I_{EO}$ & $I_C \simeq I_{CO}$

- **Forward active mode:**

  $\alpha_F I_E = 0, \alpha_R I_C \neq 0$

- **Saturation mode:**

  $\alpha_F I_E \neq 0, \alpha_R I_C = 0$

Example 10:

For a BJT, I_C = 1mA at V_{CE} = 1V, given early voltage as 75V. Calculate I_C at V_{CC} = 10V. Assume α as constant.

Sol:

$$I_c = \alpha I_0 \cdot e^{V_{BE}/\eta V_T}\left(1+\frac{V_{CE}}{V_A}\right)$$

Or $I_C = x\,(1+V_{CE}/V_A)$

Where V_A is the early voltage.

$$\therefore \frac{I_c'}{1mA} = \frac{x\left(1+\frac{10}{75}\right)}{x\left(1+\frac{1}{75}\right)}$$

$\therefore \quad I_C' = 1.118mA$

PARAMETERS OF BJT

1) Transconductance of BJT:

- Transconductance (also called mutual conductance) g_m is given as:

$$g_m = \frac{|I_c|}{V_T}\mho \qquad \text{.....1)}$$

Where

V_T is the thermal voltage (in volts)

$$V_T = \frac{KT}{q} \qquad \text{.....2)}$$

I_C is the collector current.

From 1 and 2:

$$g_m \simeq \frac{\alpha I_E}{V_T}$$

- Also, $I_C \simeq \alpha I_E$

$$\Rightarrow \quad g_m \simeq \frac{\alpha q I_E}{KT} \upsilon$$

Since,

$$\alpha \simeq 1$$

$$g_m \simeq \frac{q I_E}{KT} \upsilon$$

2) Base transit time:

$$\tau_t = \frac{W_B^2}{2D}$$

Where

- D is the diffusion constant or the coefficient of the injected minority carriers in the base.
- $D = D_n$ for NPN transistor
- $D = D_p$ for PNP transistor
- W_B is the base width of the transistor.

Example 11:

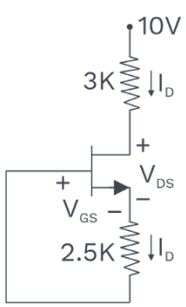
Consider the circuit in the figure below:

Assume:

I_D= 10 mA

V_P = -5 Volts

Find I_D, V_{GS}, V_{DS}, g_m



Sol:

1) $V_{GS} = -I_D R_S = -I_D \times 2.5 \times 10^3$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$\frac{-V_{GS}}{2.5 \times 10^3} = 10 \times 10^{-3}\left[1 - \frac{V_{GS}}{-5}\right]^2$$

$$-V_{GS} = 25\left[1 + \frac{V_{GS}}{25}\right]^2$$

$$= 25\frac{(25 + 10V_{GS} + V_{GS}^2)}{25}$$

$$V_{GS}^2 + 11V_{GS} + 25 = 0$$

$$V_{GS} = \frac{-11 \pm \sqrt{121 - 4 \times 1 \times 25}}{2 \times 1}$$

$V_{GS1} = -3.2$ V and

$V_{GS2} = -7.79$ V

$V_{GS2} < V_P$: Hence, this value of V_{GS} is not possible.

So, $V_{GS} = V_{GS1} = -3.2$ V

2) $I_D = \frac{-V_{GS}}{R_S} = \frac{3.2}{2.5 \times 10^3} = 1.28\,\text{mA}$

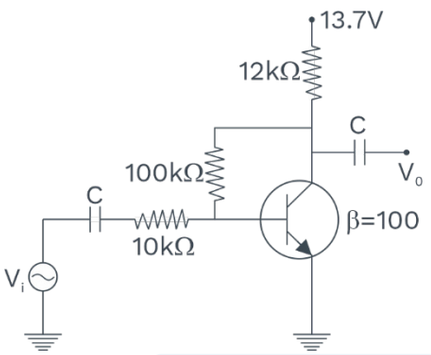
3) $V_{DS} = V_{DD} - I_D(R_D + R_S)$

$V_{DS} = 10 - 1.28[3 + 2.5] = 2.96$ V

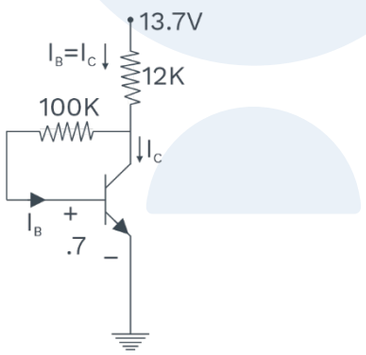
4) $g_m = \frac{2I_{DSS}}{-V_p}\left(1 - \frac{V_{GS}}{V_p}\right) = \frac{2}{-V_p}\sqrt{I_D I_{DSS}}$

$$= \frac{2 \times 10 \times 10^{-3}}{-(-5)}\left[1 - \frac{-3.2}{-5}\right] = 1.44\,\text{m}\mho$$

Example 12:
Find the value of transconductance in the circuit shown below:



Sol:
For DC analysis, consider the external capacitor to be open-circuited.
Consider the circuit below, I_C is needed to be calculated first.



By KVL in Collector-emitter loop:

$$13.7 = 12(I_B + I_C) + I_B + 0.7$$

$$13 = 12 \times 101\, I_B \times 100\, I_B$$

$$I_B = 9.90 \times 10^{-3}\ \text{mA}$$

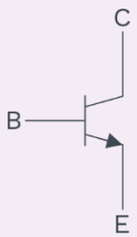
$$I_C = 0.99\text{mA}$$

$$g_m = \frac{I_C}{V_t} = \frac{0.99}{26} = 0.0380\ \text{mho}$$

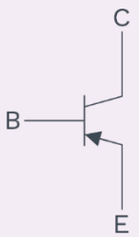
Chapter Summary

- Bipolar Junction Transistor is a three layer semiconductor device with three junctions: emitter, base, and collector. It is of 2 types:

1) NPN-BJT



2) PNP-BJT



- n- type base is made very thin as possible.
- $I_C = \alpha I_E + I_{CO}$
- $I_E = I_C + I_B$
- $\frac{i_c}{i_b} = \frac{\beta\gamma}{1-\beta\gamma} = \frac{\alpha}{1-\alpha} = \beta$
- For common emitter configuration: I_c is the output current, I_B is the input current, and V_{CE} is the output voltage.
- Common base terminology is derived from the fact that the:
  1) base is common to both input and output of the configuration.
  2) base is usually the terminal closest to or at ground potential.
- In a common collector configuration, the collector terminal is kept grounded and it is the terminal, which is common to both input, and output circuits.
- When both the emitter, and the collector functions are reverse biased. the transistor is in the cut-off region ($V_{CE} \simeq V_{CC}$).
- When the emitter junction is forward biased, and the collector junction is also forward biased. The transistor is in the saturation region ($V_{CE}=V_{CE(sat),}$).
- When the emitter junction is forward biased, and the collector junction is reverse biased.

  The condition when $V_{CE(sat)} < V_{CE} < V_{CC}$, the transistor is in the forward active region.
- When the emitter junction is reverse biased, and the collector junction is forward biased. The transistor is in the reverse active region.

- Early effect is the phenomenon when the magnitude of collector voltage increases, the effective base width of the transistor is reduced.
- The process in which by applying a larger collector voltage, the effective base width of the transistor reduces to zero is known as punch through.
- $V_{BR} \propto \frac{1}{\text{Doping concentration}}$
- **Current equations:**

1) $I_B = -\alpha_F I_E - I_{CO}\left(e^{\frac{V_E}{V_T}} - 1\right)$

2) $I_E = -\alpha_R I_C - I_{EO}\left(e^{\frac{V_E}{V_T}} - 1\right)$

3) $I_E + I_C + I_B = 0$

$\Rightarrow I_B = -(I_E + I_C)$

$$g_m \simeq \frac{q I_E}{KT} \upsilon$$

$$\tau_t = \frac{W_B^2}{2D}$$

PRACTICE QUESTIONS

Q1 The early effect in a bipolar junction transistor is caused by

a) Fast turn ON
b) Large collector-base reverse bias
c) Fast turn OFF
d) Large emitter-base forward bias

Q2 The most heavily doped region of a transistor is:

a) Base
b) Collector
c) Emitter
d) None of these

Q3 The value of $b = \frac{I_C}{I_B}$ of a BJT is:

a) > 1
b) about 0.1
c) about 10-3
d) about 10-5

Q4 The transistor, which is used for designing the digital circuits, generally has to operate in

a) Active region
b) Breakdown region
c) Cut off & Saturation region
d) All are correct

Q5 In an npn transistor, _____ are the minority carriers.

a) Free electrons
b) Acceptor ions
c) Donor ions
d) Holes

Q6 Which of the following devices belong to the family of transistors?

a) BJT and IGBT
b) SCR, GTO and BJT
c) GTO, BJT and IGBT
d) SCR and IGBT

Q7 **BJT is a three terminal device which stands for bipolar junction transistor. Which of the following are true about BJT.**

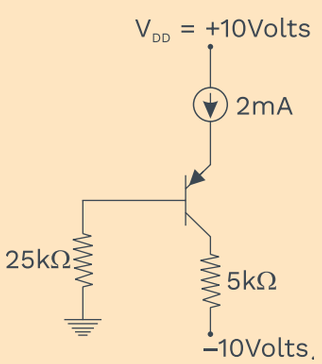
1) **Base has the smallest area to reduce the transit time.**
2) **Collector is provided with the largest area to withstand heat dissipation.**
3) **BJT is a current controlled device**
4) **BJT is a voltage controlled device**

a) 1, 2 and 3 **b) 1, 2 and 4**
c) Only 1 and 2 **d) Only 2 and 4**

Q8 **A silicon BJT has a leakage current I_{CBO} of 10 nA at 30°C. If the temperature rises to 60°C, calculate the leakage current:**

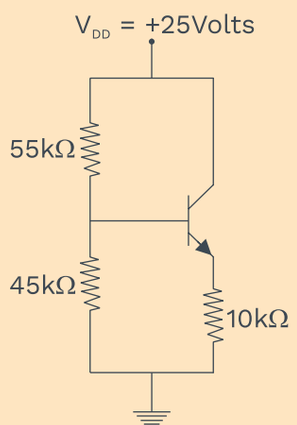
a) 80 nA **b) 40 nA**
c) 60 nA **d) 20 nA**

Q9 **What is the value of V_{EC} in the figure given below? (Assume β = 50 for the BJT connected)**



a) 6.074V **b) 1.876V**
c) 4.394V **d) 1.792V**

Q10 **What are the Q-point (I_{CQ}, V_{CEQ}) values for the circuit given in the figure below? (Let β = 100 for the BJT connected)**



a) 1.190 mA, 14.72 Volts
b) 2.019 mA, 14.52 Volts
c) 1.019 mA, 14.71 Volts
d) 2.190 mA, 14.25 Volts

Q11 **In a BJT at room temperature, what will be the voltage across the base-emitter junction if the emitter current is halved? (Let η = 1)**

a) 18 mV
b) –18mV
c) 26 mV
d) –26 mV

Answer key:

1) b)	**2) c)**	**3) a)**	**4) c)**	**5) d)**
6) a)	**7) a)**	**8) a)**	**9) b)**	**10) c)**
11) b)				